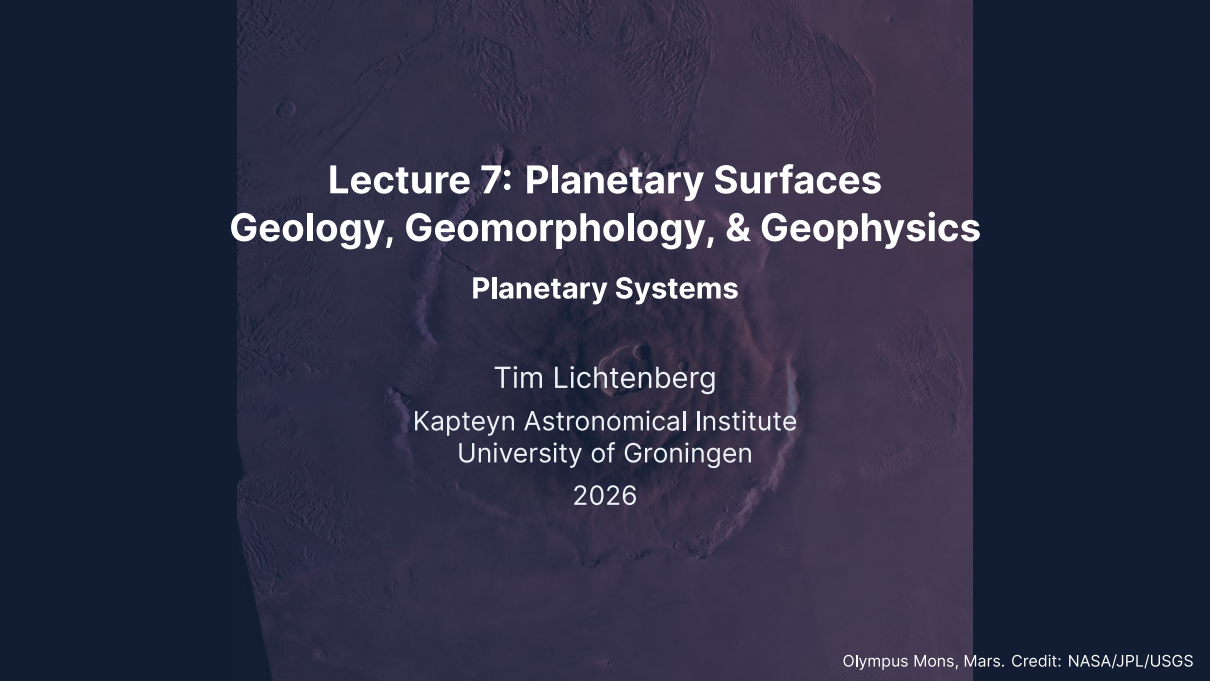


The background of the slide is a high-resolution, false-color image of the Martian surface, specifically focusing on the Olympus Mons volcano. The volcano's massive, layered rim and central caldera are visible, surrounded by a vast, desolate landscape with various smaller craters and ridges. The image is tinted with a dark, reddish-purple hue, giving it a dramatic and scientific appearance.

Lecture 7: Planetary Surfaces Geology, Geomorphology, & Geophysics

Planetary Systems

Tim Lichtenberg
Kapteyn Astronomical Institute
University of Groningen

2026

Learning objectives

By the end of this lecture, you will be able to:

- Distinguish endogenic and exogenic surface processes and the three stages of crater formation
- Derive the crater scaling law $D \sim (E/\rho g)^{1/4}$ and use crater counting for surface ages
- Compare volcanic, tectonic, and erosional styles across the solar system

Recap: Lectures 5 & 6

- Hydrostatic equilibrium and the scale height set an atmosphere's structure
- Greenhouse warming: $T_s > T_{eq}$; Clausius-Clapeyron sets where clouds form
- The carbonate-silicate cycle is Earth's long-term thermostat

Today: What shapes the *surfaces* of planets and moons?



1. Surface processes

Valles Marineris hemisphere of Mars, Viking Orbiter mosaic. Credit: NASA/JPL/USGS

Endogenic vs. exogenic processes

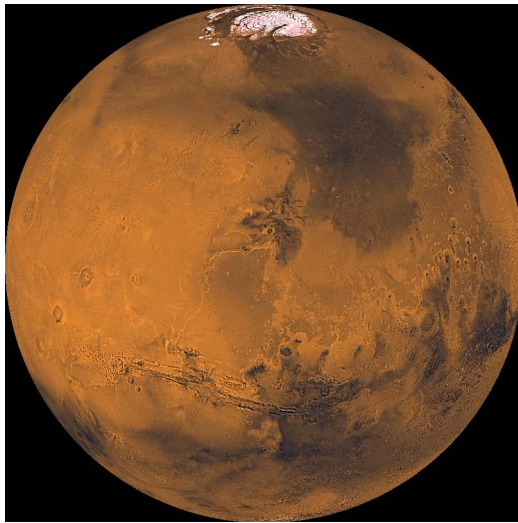
A surface is a geological record: processes that build landforms compete with processes that destroy them.

- **Endogenic** (internal heat): volcanism, tectonics (faulting, folding)
- **Exogenic** (external forces): impact cratering, erosion by wind, water and ice

Hot interior: endogenic processes win. Dead interior: the surface keeps its ancient craters.

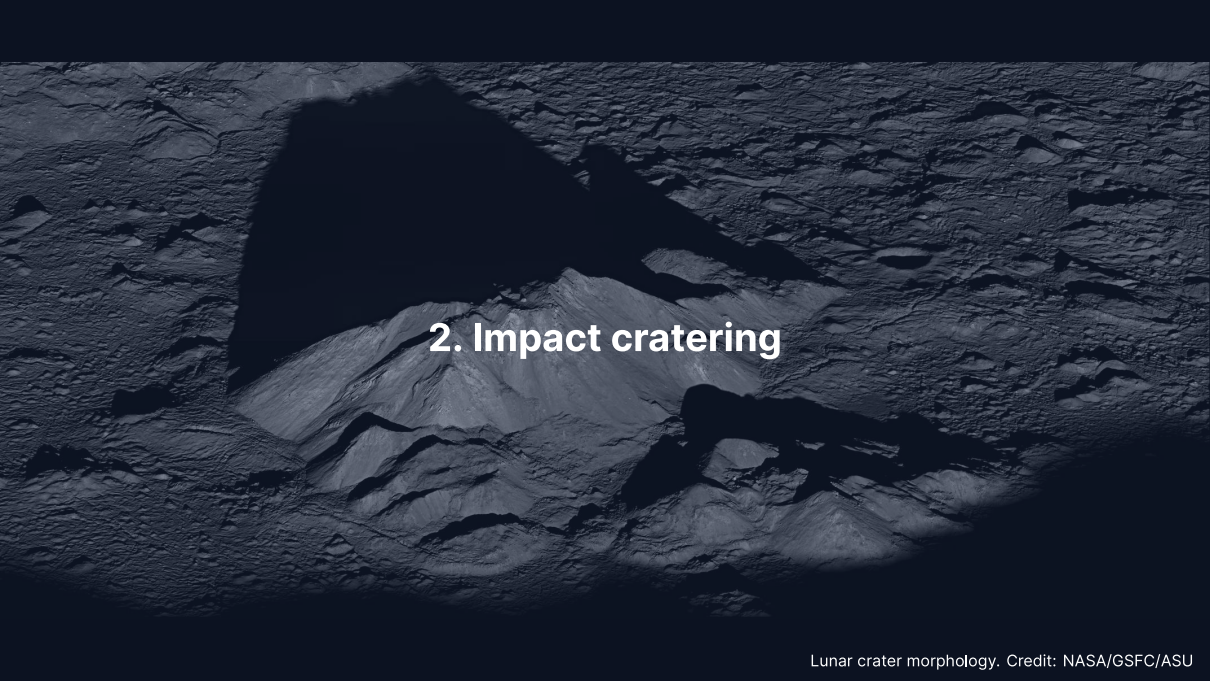
Dominant surface process by body

Body	Dominant process	Key evidence
Moon	Impact cratering	Saturated highland craters, >4 Gyr
Earth	Plate tectonics + erosion	Ocean floors <200 Myr; rapid weathering
Mars	Volcanism + impacts	Olympus Mons; cratered southern highlands
Venus	Volcanism	Volcanic plains cover ~80% of surface
Io	Tidal volcanism	~300–400 active volcanoes; age <1 Myr
Europa	Ice tectonics	Lineae, chaos terrain, very few craters



Mars shows all four surface processes on one disk: volcanism at Tharsis, rifting at Valles Marineris, cratering in the southern highlands, polar erosion. Viking Orbiter mosaic.

Credit: NASA/JPL/USGS.

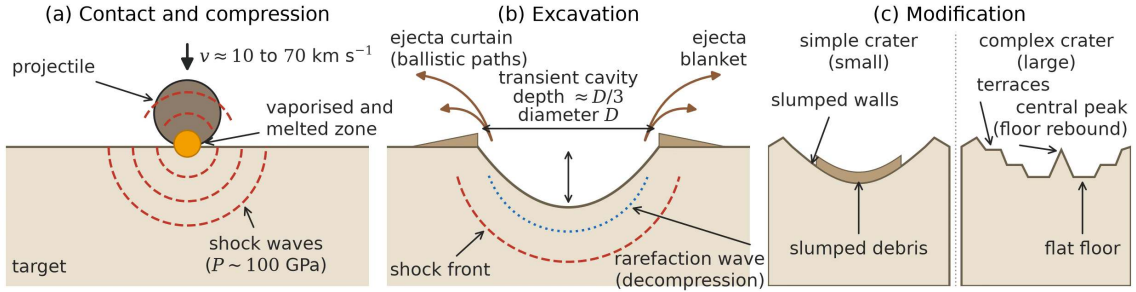
A black and white photograph of a lunar crater. The crater floor is covered in a dense field of smaller craters and rocks. A large, dark shadow is cast across the upper left portion of the image. In the center, a prominent, conical peak rises from the floor. The text "2. Impact cratering" is overlaid in white on the central peak.

2. Impact cratering

The most universal geological process

- **Every** solid body in the solar system bears impact craters
- Without atmosphere or geology (Moon, Mercury), craters survive billions of years
- Impact velocities $10\text{--}70 \text{ km s}^{-1}$, set by orbital mechanics (Lecture 2)

A 1 km rocky asteroid at 20 km s^{-1} carries $E_k = \frac{1}{2}mv^2 \approx 3 \times 10^{20} \text{ J}$, about 1000 times the largest nuclear weapon, released in under a second.



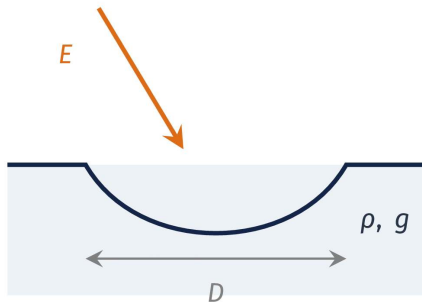
The three stages of crater formation. Course figure.

Shock near 100 GPa in under 1 s, a cavity with depth:diameter about 1:3, then a **simple crater** or a **central peak**.

3. Blackboard derivation: the crater scaling law

Caloris basin, Mercury. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington

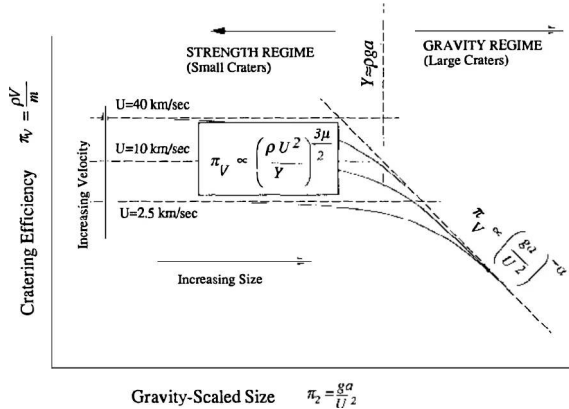
Goal and strategy



Goal: how does crater diameter D depend on impact energy E , target density ρ and gravity g ?

- **Gravity regime** (craters $\gtrsim 100$ m):
 $D = C E^a \rho^b g^c$
- Dimensions: $[E] = ML^2T^{-2}$,
 $[\rho] = ML^{-3}$, $[g] = LT^{-2}$
- Three equations for a, b, c ; then numbers

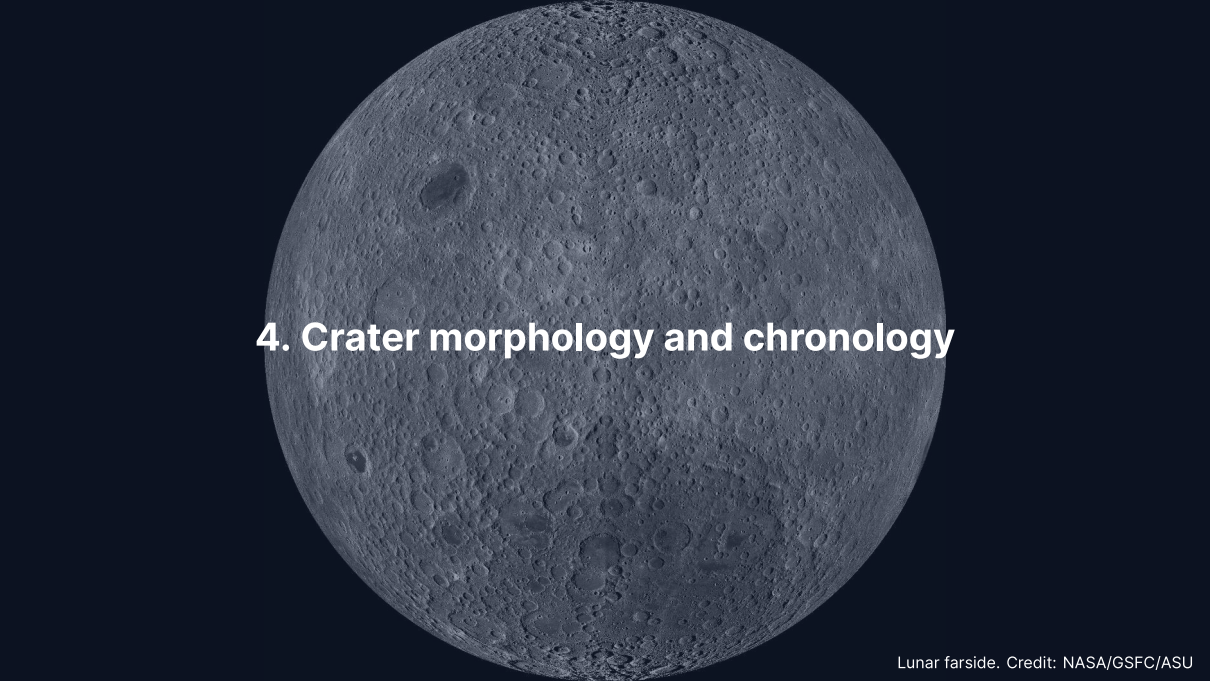
To the board.



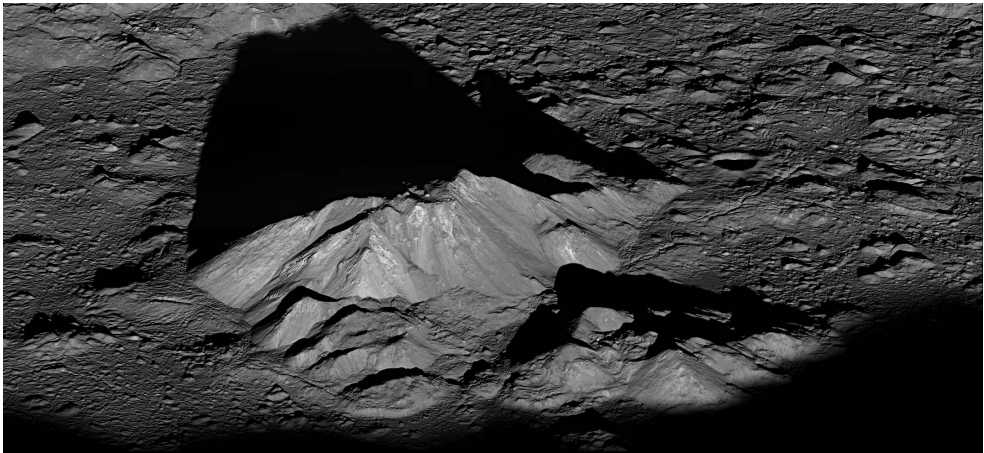
Crater size against impactor size in the strength and gravity regimes.

Credit: Holsapple (1993), Fig. 3.

$D \sim (E/\rho g)^{1/4}$ holds in the gravity regime, above about 100 m on the Moon: doubling E grows D by only $2^{1/4} \approx 1.19$, and at fixed v and ρ a 1 km impactor makes a 16.5 km crater ($D \propto L^{3/4}$).

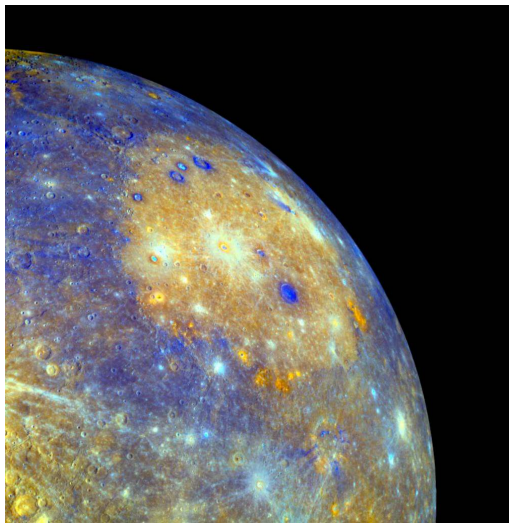


4. Crater morphology and chronology



Simple, complex and multi-ring craters on the Moon. Credit: NASA/GSFC/ASU.

Simple bowls (depth:diameter about 1:5) below $D_t \approx 15$ km on the Moon; **complex** craters with central peak, terraces and flat floor up to about 300 km; **multi-ring basins** beyond: Orientale 930, Caloris 1550, Hellas 2300 km.



Caloris basin, Mercury, about 1550 km across: concentric rings, and volcanic plains (orange) that filled it after the impact.

Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

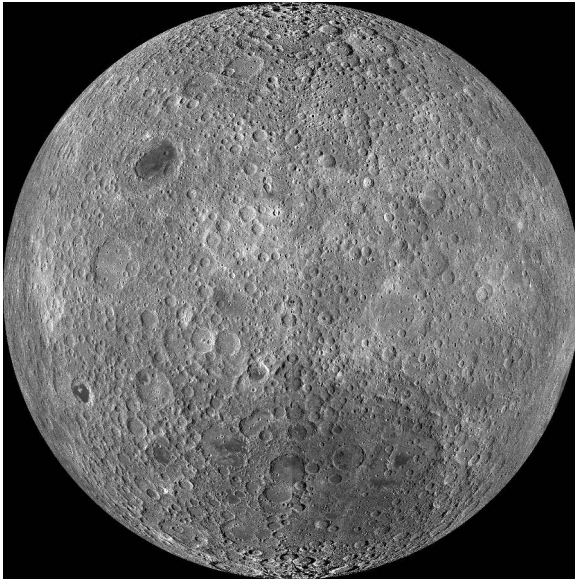
Transition diameter

The simple-to-complex transition scales with surface gravity:

$$D_t \approx D_{t,\text{Moon}} \cdot \frac{g_{\text{Moon}}}{g}$$

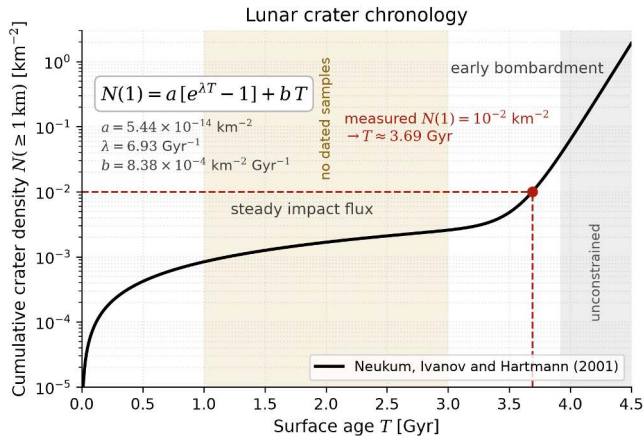
Body	$g \text{ (m s}^{-2}\text{)}$	$D_t \text{ (km)}$
Moon	1.62	15
Mars	3.71	~6.5
Earth	9.81	~2.5
Mercury	3.70	~6.6

On Earth, virtually all craters larger than a few km are complex.



Crater counting on the lunar farside: older surfaces carry more craters. Credit: NASA/GSFC/ASU; method after Neukum et al. (2001).

$N(> D) = a D^{-b}$ with $b \approx 2-3$, a straight line on log-log axes: a higher line is an older surface, so counts give **relative** ages until **saturation**, and Apollo and Luna samples supply the **absolute** calibration.



Lunar cratering chronology $N(1) = a[e^{\lambda T} - 1] + bT$. Course figure after Neukum et al. (2001).

The linear term is the steady flux of the last 3 Gyr, the exponential term the early bombardment before 3.6 Gyr; $N(1) = 10^{-2} \text{ km}^{-2}$ reads as 3.7 Gyr, and the shaded bands hold no dated sample: interpolation, not calibration.

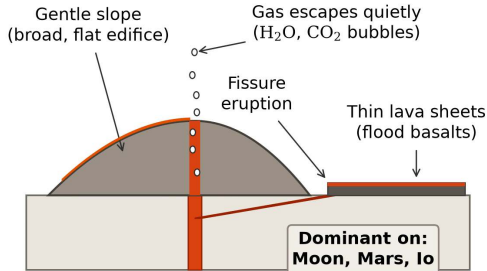


5. Volcanism

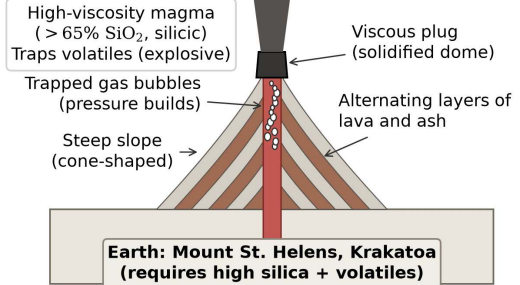
Olympus Mons, Mars. Credit: NASA/JPL/USGS

(a) Effusive volcanism (schematic)

Low-viscosity magma
(~ 50% SiO₂, basaltic)
Flows easily (effusive)

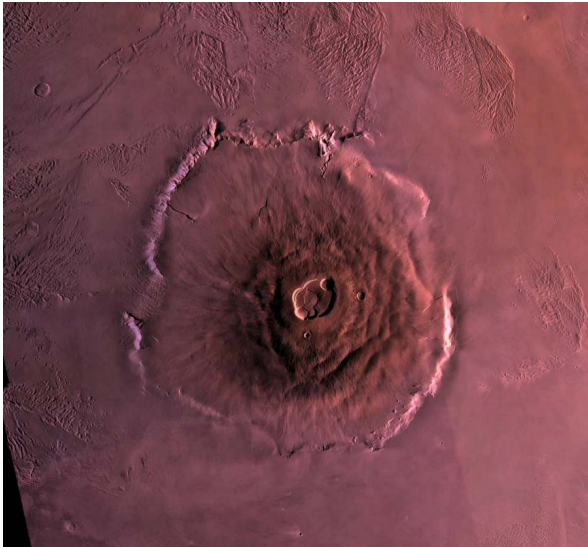


(b) Explosive volcanism (schematic) Explosive eruption column (ash and gas plume)

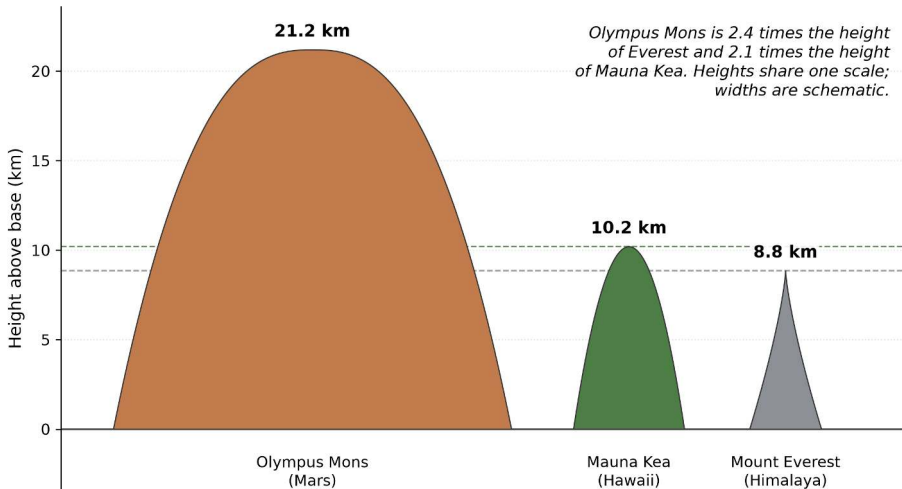


Effusive against explosive volcanism. Course figure.

Basaltic magma (about 50% SiO₂) flows: **effusive** shields and flood basalts; silicic magma (>65%) traps volatiles: **explosive** eruptions.



Olympus Mons, Mars: a shield volcano about 600 km wide and 21 km high, with nested summit calderas. Viking Orbiter mosaic. Credit: NASA/JPL/USGS.



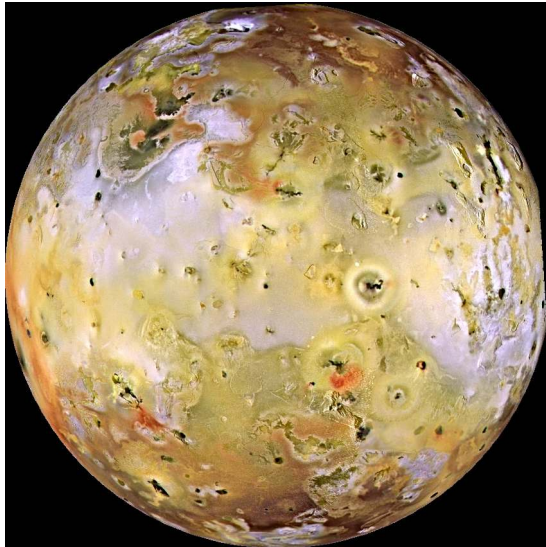
Olympus Mons, Mauna Kea and Everest on one height scale, widths schematic. Course figure.

No plate motion, $g \approx 0.38 g_{\oplus}$: one plume, one edifice for Gyr; a pile about $2.6\times$ taller.

Volcanism across the solar system

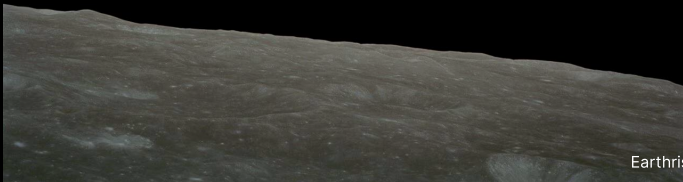
Body	Style	Driving mechanism	Examples
Earth	Effusive + explosive	Internal heat + recycling	Hawaii, mid-ocean ridges
Moon	Flood basalts (extinct)	Residual heat (3.9–3.1 Ga)	Maria
Mars	Shield volcanoes	Mantle plumes, no plates	Olympus Mons, Tharsis
Venus	Lava plains + shields	Internal heat, stagnant lid	Maat Mons
Io	Ultramafic flows	Tidal heating	Loki Patera, Pele

Venus returns in Lecture 9, Io in Lecture 11.



Io from Galileo SSI: active sulfur and silicate volcanism powered by tidal dissipation in the Laplace resonance. Credit: NASA/JPL-Caltech/University of Arizona.

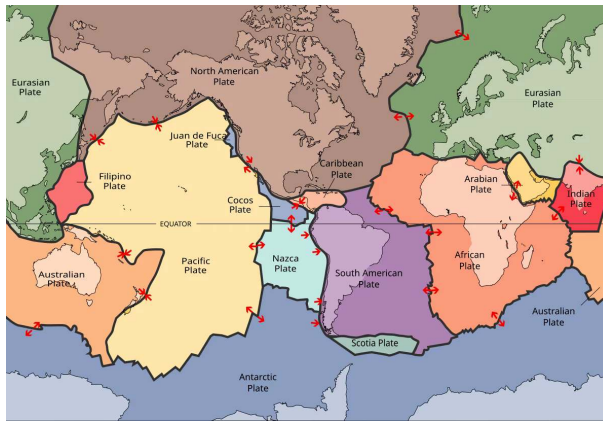
Break



An aerial photograph of the Carrizo Plain in California, showing the San Andreas Fault as a prominent, winding linear depression. The landscape is arid and flat, with the fault line clearly visible as a series of connected ridges and troughs. The text "6. Tectonics" is overlaid in the center of the image.

6. Tectonics

The San Andreas Fault across the Carrizo Plain, California. Credit: Ikluft, Wikimedia Commons, CC BY-SA 4.0



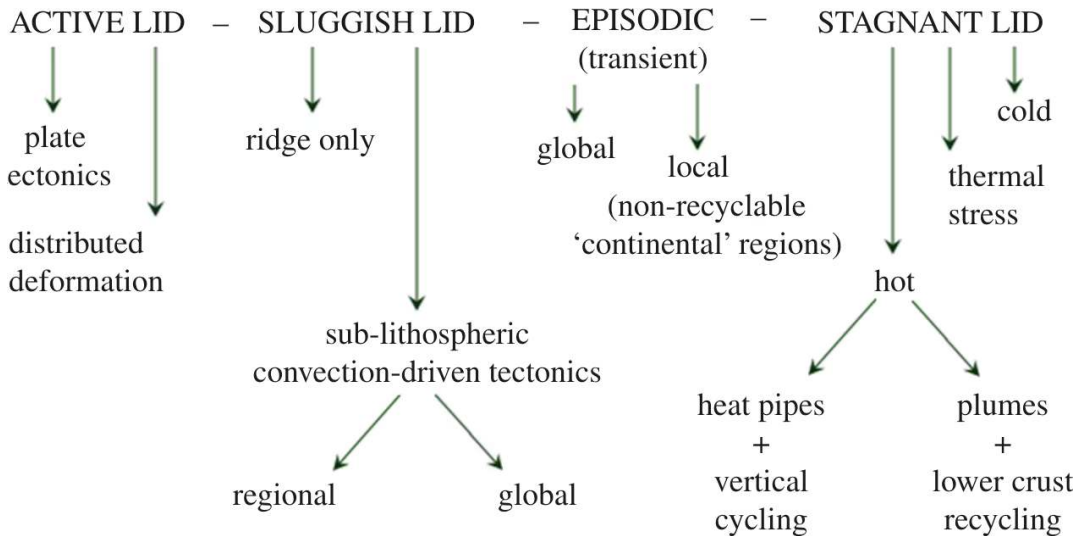
The tectonic plates of Earth and their boundaries. Credit: Wikimedia, CC BY-SA 3.0.

Earth is the **only** body with active plate tectonics: about 15 rigid plates at $1\text{--}10\text{ cm yr}^{-1}$, **divergent** ridges that make crust, **convergent** subduction that recycles it, **transform** faults that slide; subduction closes the **carbonate-silicate cycle**.

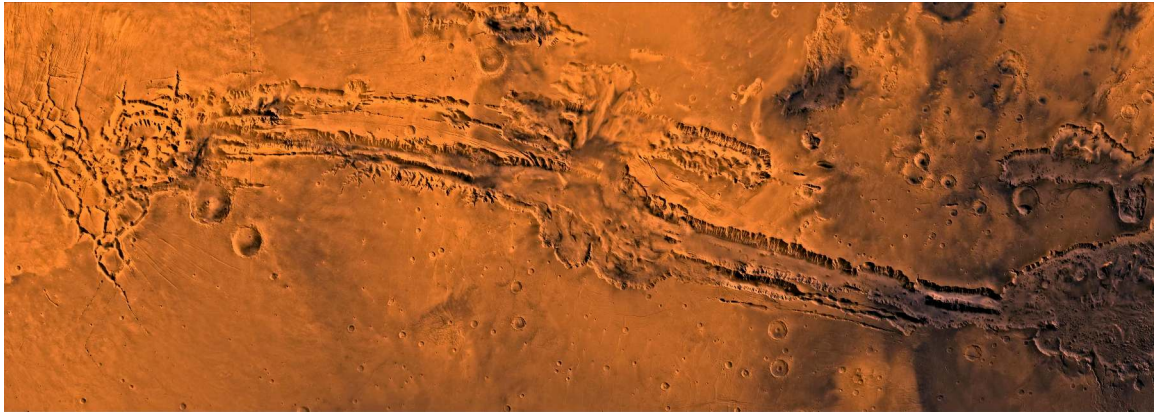
Stagnant-lid convection

- Every other terrestrial body: one rigid lithospheric lid over a convecting mantle
- Heat leaves by conduction and occasional volcanism; the thick lid keeps the mantle warm
- Melt reaches the surface only through rare plumes (Tharsis), and volcanism fades as the lid grows

Stagnant lid is the default for temperature-dependent viscosity. Breaking the lid, as Earth does, needs water weakening and self-sustaining damage.

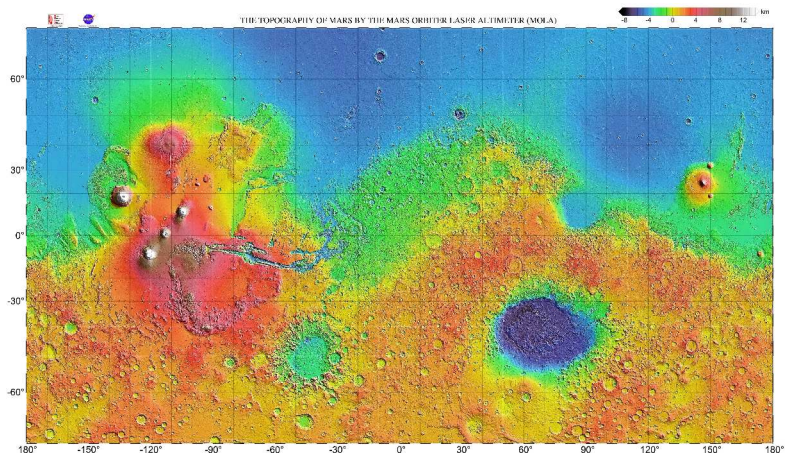


Tectonic modes of terrestrial planets beyond the mobile-stagnant binary: active lid, sluggish lid, episodic regimes, and stagnant lid (cold or hot). Credit: Lenardic (2018), Fig. 2.



Valles Marineris in a Viking Orbiter mosaic, about 4000 km long and up to 7 km deep: no plate boundaries, yet the largest rift in the solar system, opened by the Tharsis volcanic load.

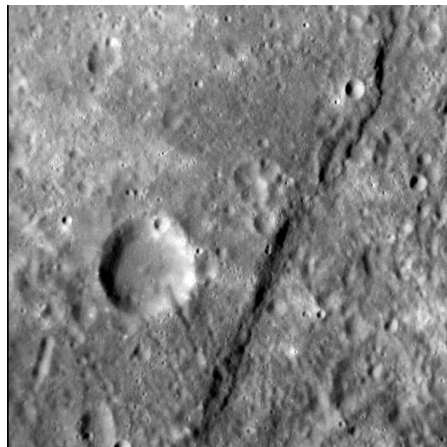
Credit: NASA/JPL/USGS.



MOLA topography of Mars: the hemispheric dichotomy.

Credit: NASA/JPL/GSFC/MOLA Science Team.

Highlands about 6 km above the lowlands; four giant volcanoes on Tharsis, Hellas at -8.2 km, Valles Marineris on the equator.



A lobate scarp on Mercury, a thrust fault that cuts across craters, MESSENGER.

Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

The contraction is younger than the craters it cuts: the cooling iron core shrank Mercury's radius by up to about 7 km.



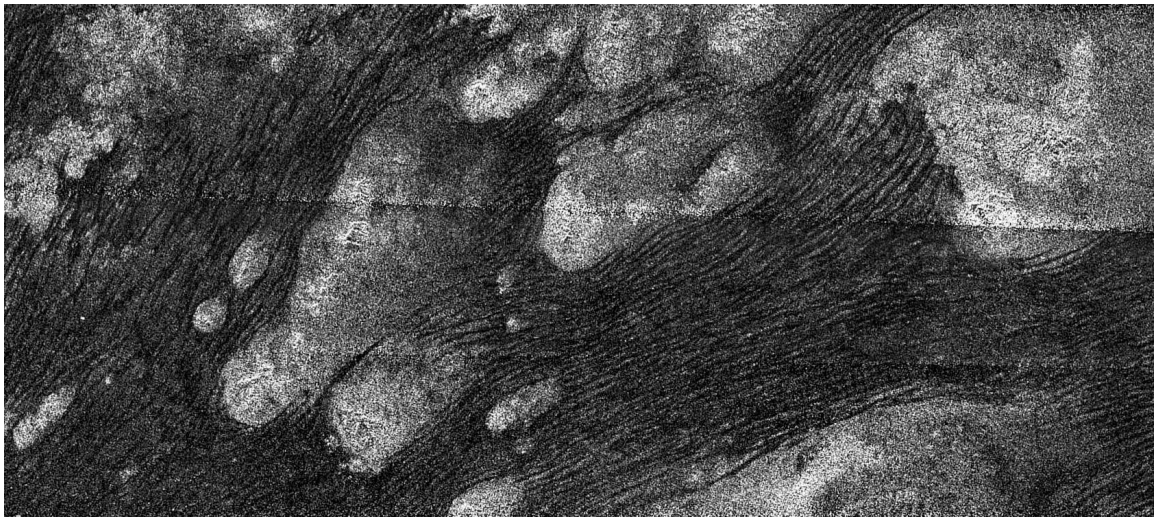
7. Erosion and weathering

Barchan dunes in Nili Patera, Mars, imaged by HiRISE. Credit: NASA/JPL-Caltech/University of Arizona

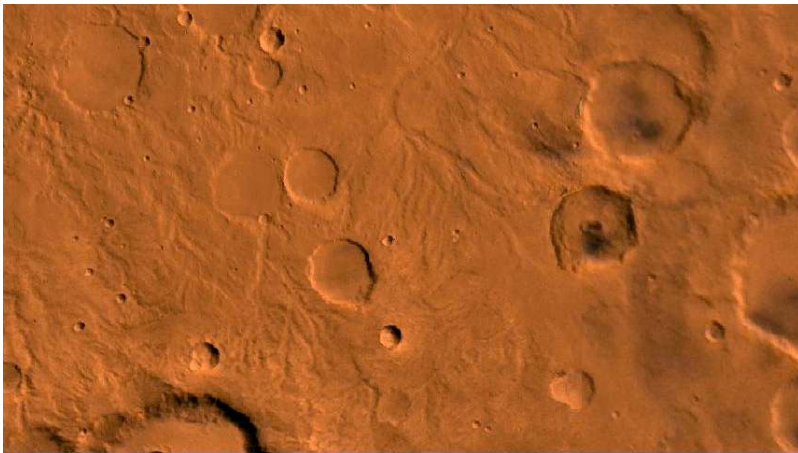


Barchan dunes in Nili Patera, Mars, HiRISE: repeat images show whole dunes advancing at Earth-like sand fluxes under an atmosphere 170 times thinner (Bridges et al. 2012).

Credit: NASA/JPL-Caltech/University of Arizona.



Longitudinal dunes in Titan's Shangri-La region, Cassini SAR: organic sand, water-ice bedrock, the same aeolian physics. Credit: NASA/JPL-Caltech/ASI.

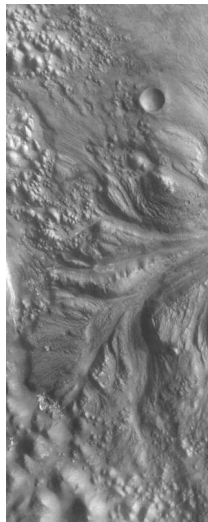


Dendritic valley networks in the Noachian highlands of Mars. Credit: NASA/JPL/USGS.

Mars valley networks (>3.7 Ga) record sustained liquid water, **outflow channels** hundreds of km long record catastrophic groundwater release, and on **Titan** methane rivers rounded the ice pebbles Huygens landed on in 2005.

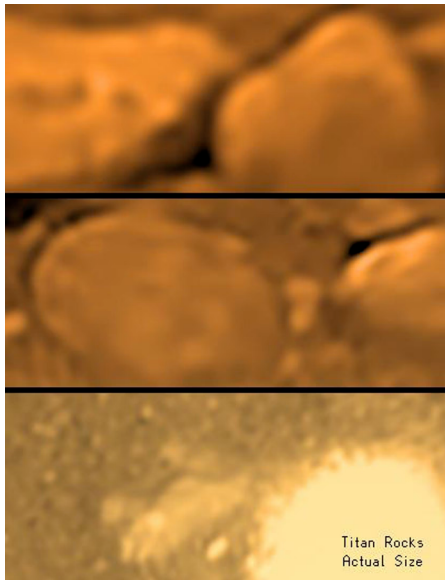


The Colorado River in the Grand Canyon has cut about 1.6 km through flat-lying Paleozoic sediments in 5 to 6 Myr, the type example of sustained fluvial erosion; resistant limestone stands as cliffs, weaker shale erodes to slopes. Credit: Alex Demas/USGS, public domain.



An outflow channel through Aram Chaos: braided islands and scoured troughs record catastrophic floods, in contrast to the dendritic valley networks that record precipitation.

Credit: NASA/JPL-Caltech/MSSS.



The surface of Titan from ESA's Huygens probe, 14 January 2005, in three enlarged crops. Credit: ESA/NASA/JPL-Caltech/University of Arizona, public domain.

The only image from the surface of an outer-solar-system body: decimetre blocks of water ice, rounded by methane fluvial transport, on a damp plain of organic sediment in a dried-out riverbed; liquid methane shapes Titan today.

Glacial and chemical weathering



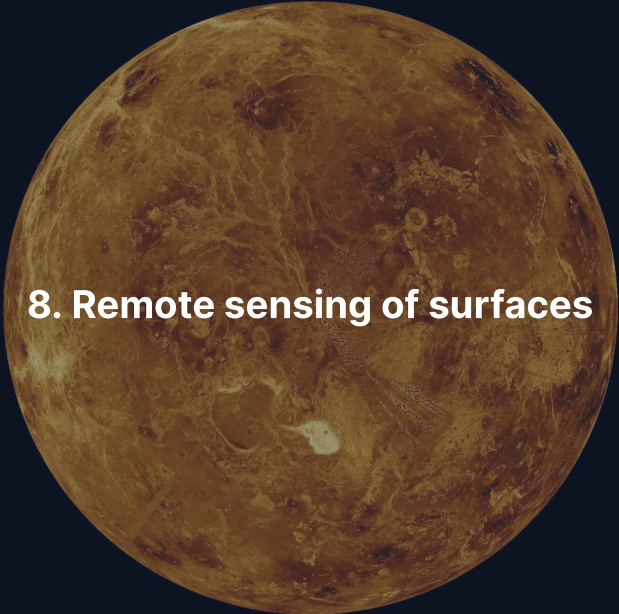
Fedchenko Glacier, Pamir. Credit: NASA/ISS



Clay-bearing unit, Glen Torridon, Gale crater. Credit:

NASA/JPL-Caltech/MSSS

- **Glacial:** ice flows under its own weight; Earth's ice sheets covered about 30% of land at the Last Glacial Maximum, Mars has polar caps and buried mid-latitude ice
- **Chemical:** silicate weathering is Earth's carbon sink; on Mars, clays, sulfates and carbonates seen from orbit record past liquid water



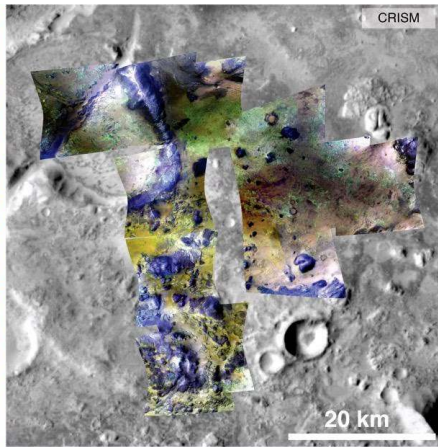
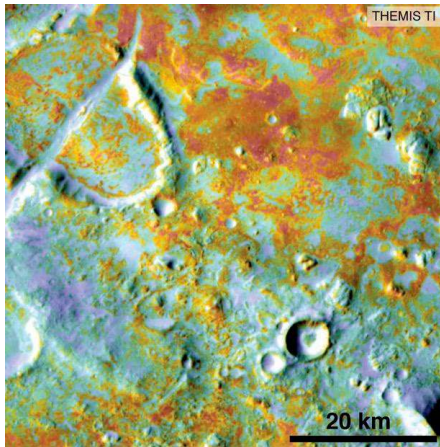
8. Remote sensing of surfaces

Venus in Magellan radar, the surface seen through an opaque atmosphere. Credit: NASA/JPL



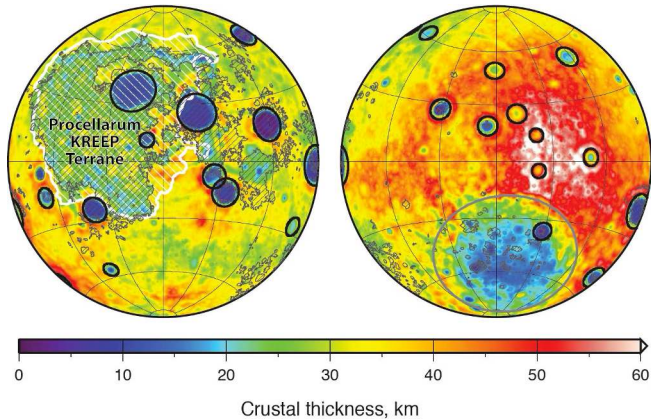
Venus from the Magellan radar mission (1990 to 1994), coloured from Venera lander measurements. Credit: NASA/JPL, public domain.

Radar imaging sees through clouds and haze: 98% of Venus mapped at about 100 m resolution, more than 1600 volcanic centres, lava plains over about 80% of the surface, and a uniform crater population that implies a mean surface age of only 300 to 700 Myr.



Nili Fossae: thermal inertia (left) and CRISM minerals (right), Mg-carbonates (green), olivine (yellow), basalt (blue). Credit: NASA/JPL-Caltech/ASU/JHU APL.

Reflectance spectroscopy reads minerals from absorption bands (CRISM: clays, carbonates); we land at a few sites, orbit gives the rest.



Lunar crustal thickness from GRAIL, nearside (left) and farside (right).

Credit: NASA/MIT/GSFC/JPL-Caltech.

Gravity field mapping reveals buried mass: the crust is 34–43 km on average and thin under the big basins, so a basin is a hole through most of the crust; **laser altimetry** (MOLA) supplies the topography to about 1 m.



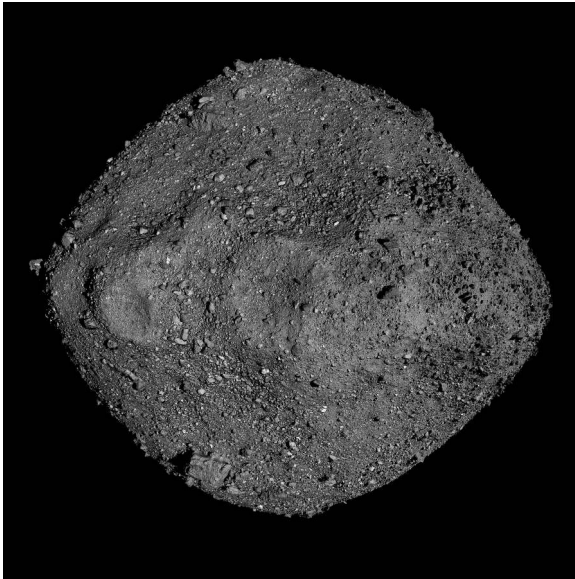
9. Regolith and space weathering

Apollo 11 bootprint in the lunar regolith. Credit: NASA



Apollo 11 bootprint in the lunar regolith. Credit: NASA/Apollo 11.

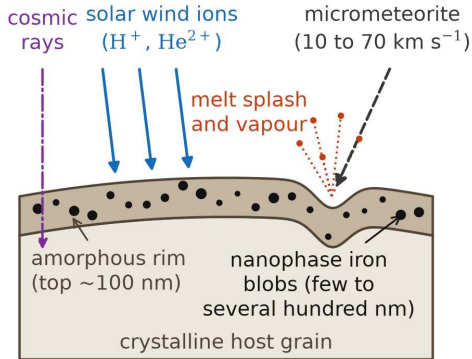
Regolith: loose debris made by impact gardening, 2–4 m on the maria and 6–8 m on the highlands; **space weathering** by solar wind and micrometeorites makes nanophase iron, so surfaces turn **darker and redder** and fresh craters like Tycho look bright.



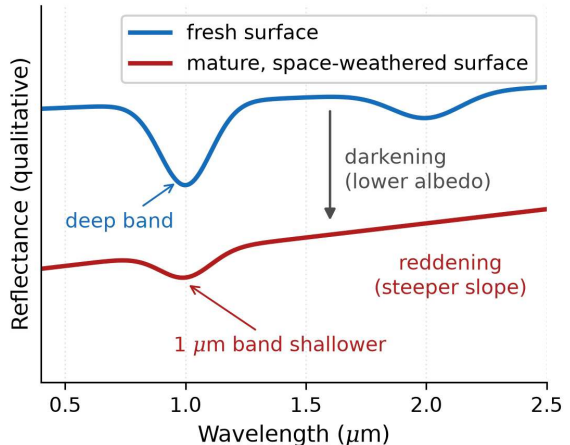
Bennu from OSIRIS-REx in December 2018, from 24 km: boulders at every scale on a body about 490 m across. Credit: NASA/Goddard/University of Arizona, public domain.

No smooth ponded plains and a very low bulk density, so high porosity: a rubble pile of collisional fragments reaccumulated under self-gravity; OSIRIS-REx returned 121.6 g of it to Earth in September 2023.

(a) Regolith grain cross-section (schematic)



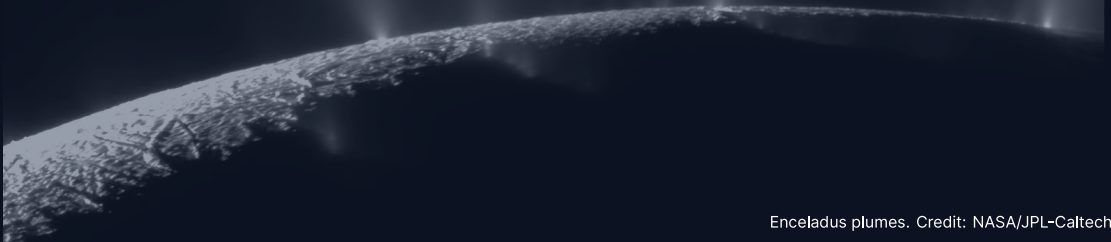
(b) Reflectance spectra (schematic)

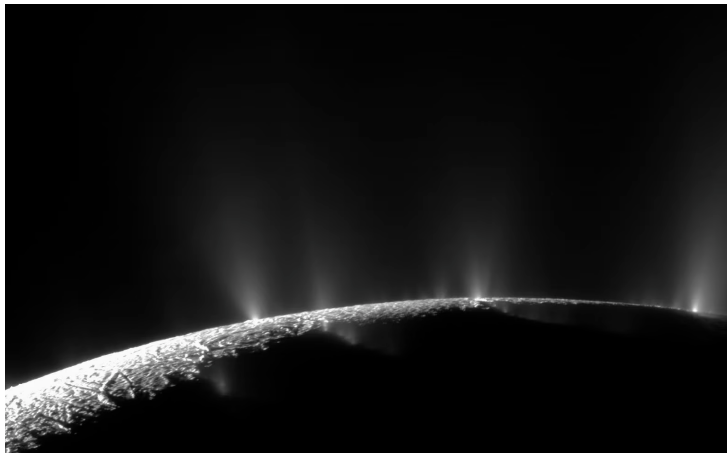


Space weathering: solar wind ions and micrometeorites build an amorphous rim with nanophase iron; mature surfaces are darker, redder and show a shallower 1 μm band.

Course figure.

10. Cryovolcanism on icy bodies





Water plumes from the south pole of Enceladus, Cassini. Credit: NASA/JPL-Caltech/SSI.

The “magma” is water, ammonia-water or methane at 100–270 K (nitrogen gas at about 38 K on Triton); **tidal heating** keeps oceans liquid far outside the habitable zone: Enceladus erupts, Europa resurfaces, Triton vents (Lecture 11).



Double ridges and lineae near Pwyll crater on Europa, Galileo SSI: so few craters that the surface is only 40–90 Myr old.

Credit: NASA/JPL-Caltech/University of Arizona/University of Colorado.



Voyager 2 mosaic of Triton: nitrogen geyser streaks cross the pink N_2/CH_4 polar frost (bottom), cantaloupe terrain of dimples and ridges fills the north.

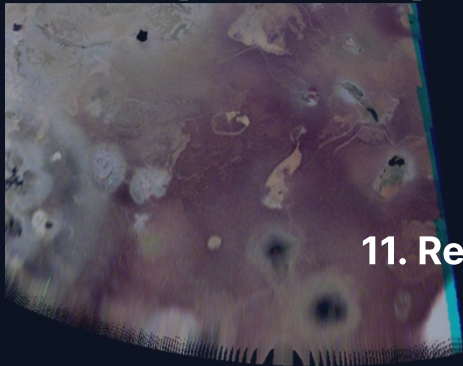
Credit: NASA/JPL/Voyager 2.



Pluto in enhanced true colour from New Horizons, 14 July 2015: Sputnik Planitia is the bright western lobe of the heart-shaped Tombaugh Regio. Credit: NASA/JHUAPL/SwRI, public domain.

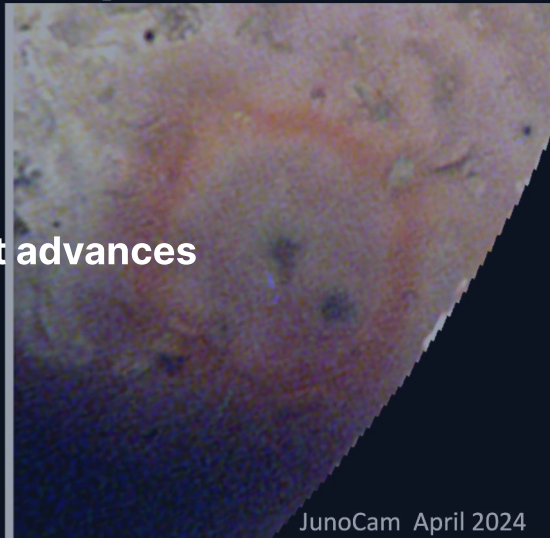
A crater-free plain of convecting nitrogen ice bordered by water-ice mountains, and dark reddish terrains coated in tholins from radiation chemistry: a small, distant dwarf planet can be geologically active today.

Change: New Bright Red Ring at Nusku



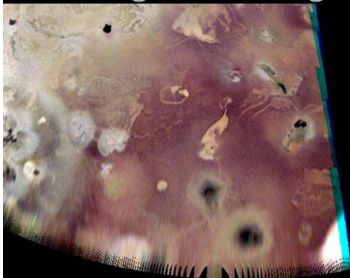
JunoCam February 2024

11. Recent advances

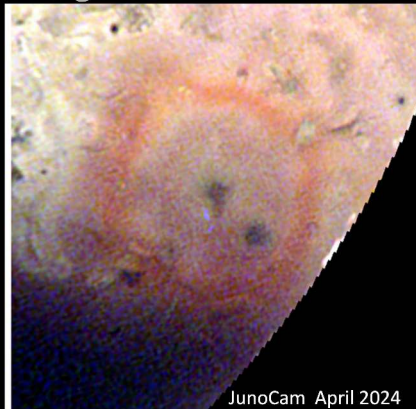


JunoCam April 2024

Change: New Bright Red Ring at Nusku



JunoCam February 2024



JunoCam April 2024

The Nusku region on Io in February and April 2024, JunoCam: a red ring of fresh sulfur-rich pyroclastic deposits appears between two flybys, resurfacing visible within months.

Credit: NASA/JPL-Caltech/SwRI/MSSS/Jason Perry.

Summary

1. Surfaces record **endogenic** (volcanism, tectonics) against **exogenic** (impacts, erosion) processes; crater counting plus Apollo calibration gives absolute ages
2. Crater scaling law: $D \sim (E/\rho g)^{1/4}$, a fourth root of the energy
3. No plate tectonics: giant volcanoes, one rift, a shrinking Mercury; Earth alone breaks its lid, and fluvial features on Mars are strong evidence for past liquid water

Next: Lecture 8 goes underneath, to the interiors that drive the endogenic half of this record.

Questions?

Next: Lecture 8, Planetary interiors: structure, composition & dynamics

Thu 1 Oct, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 2 Oct, 13:00–15:00, room 5419.0161 (Collegezaal)

<https://ips.formingworlds.space>